

A Lightweight NMEA 0183 Parser for Field Validation of a Wearable GNSS Subsystem

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Abstract—This report presents the design and field evaluation of a lightweight, dependency-free NMEA 0183 parser developed to support validation of the GNSS subsystem of a Screenless Smart Wristband under development at EDABK Lab, Hanoi University of Science and Technology. The wristband, built around a Nordic nRF52840 microcontroller and a Quectel LC76G multi-constellation GNSS module, has no display, so raw GNSS output is the primary signal available for debugging position and fix quality during field testing. This output is dense, abbreviation-heavy, and interleaved across up to five satellite constellations, which makes manual inspection slow and error-prone. The parser described here ingests timestamped raw log files, isolates and tokenizes individual NMEA sentences, converts coordinates from the NMEA degree-minute format to signed decimal degrees, and produces a structured, per-sentence-type, human-readable report together with running statistics. I describe the parsing pipeline, the dispatch logic for six core sentence types (GGA, RMC, GSA, GSV, VTG, GLL), and the program’s behavior on malformed or incomplete input. Evaluation on a 131-second field log containing 1300 parsed NMEA sentences across four GNSS constellations confirms correct decoding of position, satellite-count, and fix-quality fields, and demonstrates graceful handling of non-NMEA content – including a dropped-byte fragment and an unplanned mid-capture reboot of the GNSS module – without crashing or producing corrupted output. The tool is intended as a low-overhead diagnostic aid for ongoing GNSS validation work on the wristband project.

Index Terms—NMEA 0183, GNSS, embedded systems, log parsing, wearable devices, multi-constellation positioning

I. INTRODUCTION

The Screenless Smart Wristband is a senior graduation project at EDABK Lab, HUST, built around a Nordic nRF52840 microcontroller and a GNSS positioning module. Because the device has no display, all outdoor-positioning debugging during integration and field testing must be done by inspecting the GNSS module’s raw serial output rather than by glancing at an on-device readout. This places an unusual amount of weight on log inspection, where any tool that makes the raw log easier to read directly speeds up hardware validation.

The GNSS module used on the wristband, a Quectel LC76G, communicates over UART using the NMEA 0183 protocol, an ASCII sentence-based format originally designed

for marine electronics and now used almost universally by GNSS receivers. A single second of operation can produce a dozen or more interleaved sentences spanning GPS, GLONASS, Galileo, and BeiDou, each with a different field layout. Reading this stream by eye to confirm, for example, that the fix quality is acceptable or that a reasonable number of satellites are being tracked, is tedious and easy to get wrong.

This report describes a small, standard-library-only C program written to address that problem. The program is not part of the wristband firmware itself; it is an offline diagnostic tool that reads a timestamped capture of the module’s raw NMEA output and produces a structured, readable report, with running counts of every sentence type seen. My contribution, and the scope of this report, is limited to this parsing and validation tool. The wristband hardware and firmware are the work of the EDABK Lab senior project team and are described here only as context for why the parser was needed.

The remainder of this report is organized as follows. Section II reviews the NMEA 0183 protocol, GNSS constellations, and the LC76G module. Section III describes the parser’s design and implementation. Section IV evaluates the parser on a real field log captured from the wristband’s GNSS module. Section V concludes.

II. BACKGROUND

A. The NMEA 0183 Protocol

NMEA 0183 is an ASCII, single-talker/multiple-listener serial protocol maintained by the National Marine Electronics Association [1]. Each message, or “sentence,” begins with a \$ character, followed by a two-letter talker identifier and a three-letter sentence formatter, followed by a comma-delimited sequence of data fields, and terminated by an optional checksum field introduced by a * delimiter. Because fields may be empty (e.g., when no fix is available), a parser must locate fields by counting delimiters rather than by fixed character offsets. Although the protocol predates GPS, its simplicity over a single UART pin made it the de facto standard for GNSS receivers, including embedded modules with no other interface budget to spare.

B. GNSS Constellations and Talker Identifiers

Modern GNSS receivers can track signals from more than one satellite constellation: GPS (United States), GLONASS

I am a high school student at The Dewey Schools Tay Ho (Adventure International Program). This work was carried out as part of a lab-assistant contribution to EDABK Lab beginning April 2026.

(Russia), Galileo (European Union), and BeiDou (China). NMEA 0183 identifies the source constellation of a sentence through its two-letter talker ID: GP for GPS, GL for GLONASS, GA for Galileo, and GB for BeiDou. A receiver that fuses observations from multiple constellations into a single position solution typically labels that combined output with the talker ID GN. Multi-constellation operation generally improves time-to-first-fix and positioning accuracy, particularly in partially obstructed sky views, which is relevant to a wrist-worn device that will be used outdoors under varying conditions.

C. The Quectel LC76G GNSS Module

The Quectel LC76G is a compact, low-power GNSS module capable of concurrently tracking GPS, GLONASS, BeiDou, Galileo, and QZSS signals [?], [3]. Its small footprint and low power draw make it a common choice for battery-powered wearables, which is why it was selected for the wristband. The module outputs standard NMEA 0183 sentences over UART, including the six sentence types handled by the parser described in this report.

D. Why Structured Parsing Matters for Wearable GNSS Validation

During field testing, the wristband team needs to answer simple questions quickly: is the module producing a valid fix, how many satellites are visible per constellation, and is positioning accuracy reasonable for a near-static test point. Each of these answers is, in principle, present in the raw log, but scattered across dozens of abbreviated, comma-delimited sentences per second, several of which (GSA, GSV) repeat once per constellation per epoch. Because the device is screenless, there is no simpler on-device alternative to this kind of offline log inspection. A tool that decodes coordinates, labels fields, and tallies sentence counts turns a multi-thousand-line ASCII dump into something that can be scanned in seconds rather than read line by line.

III. SYSTEM DESIGN AND IMPLEMENTATION

A. Pipeline Overview

The parser is implemented as a single-pass C program with six logical stages, shown in Fig. 1. A raw log file is read line by line; each line is checked for a leading timestamp and an NMEA sentence; the sentence is tokenized into fields; the three-letter sentence formatter is used to dispatch to a per-type handler; the handler decodes and writes a structured block to the output report; and a set of running counters is updated for the final summary table.

B. Timestamp and Sentence Extraction

Each line in the captured log is prefixed with a bracketed timestamp, [YYYY-MM-DD_HH:MM:SS:mmm], immediately followed by the NMEA sentence. The function `extract_parts` copies the bracketed timestamp into a separate buffer and returns a pointer to the \$ character that begins the sentence. If a line begins directly with \$ (no

timestamp), it is still handled correctly. If no \$ character is found at all, the function returns NULL and the caller writes the line verbatim under a “non-NMEA” label rather than guessing at its structure. This fallback path is what allows the program to tolerate corrupted or non-sentence lines, discussed further in Section IV.

C. Tokenization

NMEA fields are separated by commas, and the sentence ends at a checksum delimiter, *, after which the remaining characters are a hexadecimal checksum rather than data. The tokenizer, `split_csv`, scans for the nearer of the next comma or * on each iteration, copies the field into a fixed 64-byte buffer with explicit null-termination, and stops as soon as a * is reached. The checksum itself is never validated; * is used only as a tokenization boundary, so a sentence with a corrupted body but an intact structure would still be parsed and reported like normal. Each sentence handler allocates a fields array sized for the maximum field count of that sentence type (e.g., 24 for GSV, which can carry up to four full satellite records per message), which bounds stack usage per call.

D. Coordinate Conversion

NMEA latitude and longitude are encoded as `ddmm.mmmm` (degrees followed directly by minutes), rather than decimal degrees. The conversion function `nmea_to_decimal` recovers decimal degrees as

$$\text{decimal} = \left\lfloor \frac{\text{raw}}{100} \right\rfloor + \frac{\text{raw} \bmod 100}{60}, \quad (1)$$

and negates the result when the hemisphere field is S or W. For example, a raw GGA field of 2055.281683 N is converted to $20 + 55.281683/60 = 20.921361^\circ\text{N}$, matching the field as logged.

E. Talker ID to Constellation Mapping

The function `constellation_name` maps the two-letter talker ID prefix of a sentence (\$GP, \$GL, \$GA, \$GB, \$GN) to a human-readable constellation name (GPS, GLONASS, Galileo, BeiDou, or Multi-constellation), used throughout the structured output so that, for example, a GSV block clearly states which constellation’s satellites it describes.

F. Per-Sentence-Type Handlers

Six dedicated handler functions decode the sentence types observed in the field log:

- **GGA** – the primary fix sentence: UTC time, latitude/longitude, numeric fix quality (mapped to a label such as “GPS fix” or “RTK fixed”), satellite count, HDOP, altitude, and geoid separation.
- **RMC** – the only standard sentence carrying date alongside time; also reports an active/void status flag, speed (converted from knots to km/h), and course. Position and speed fields are only decoded when the status flag indicates an active fix.
- **GSA** – active satellite PRNs and dilution-of-precision (DOP) values. Each constellation reports its own GSA

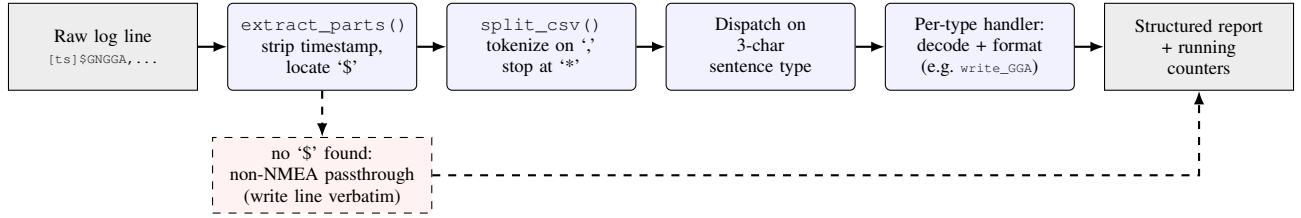


Fig. 1. Parsing pipeline. Each raw log line passes through timestamp/sentence extraction, comma tokenization, type dispatch, and a dedicated decoder before being written to the structured report. Lines with no \$ character bypass tokenization entirely and are written verbatim under a non-NMEA label (dashed path).

sentence, so up to four GSA sentences are expected per epoch in a four-constellation receiver.

- **GSV** – satellites in view, in PRN/elevation/azimuth/SNR quadruples. A single epoch’s satellites for one constellation may be split across multiple GSV messages (up to four satellites per message); the handler tracks a running satellite index across messages.
- **VTG** – course over ground and speed, reported in both knots and km/h; redundant with RMC’s speed/course fields but useful as a cross-check.
- **GLL** – a legacy latitude/longitude sentence inherited from older LORAN-era equipment; included for completeness, though it carries strictly less information than GGA.

Every handler begins with a field-count guard: if a sentence has fewer fields than its type requires, the handler writes an explicit “(incomplete sentence, skipped)” note instead of indexing past the end of the tokenized array. Sentences that do not match any of the six known formatters (proprietary \$PAIR/\$PQTM messages, or other vendor-specific output) are written verbatim to the report rather than discarded, so that no information is silently lost.

G. Dispatch Loop and Robustness

The main loop reads one line at a time, extracts the timestamp and sentence, and dispatches on the three characters immediately following the two-letter talker ID – a technique that lets the same dispatch logic handle a sentence regardless of which constellation produced it (e.g., \$GPGSV and \$GLGSV both dispatch to `write_GSV`). Seven running counters (one per known sentence type plus one for everything else) are incremented as each line is processed and are printed as a summary table once the file is exhausted. Because all string handling uses fixed-size buffers with explicit length checks and null-termination, malformed input cannot overrun a buffer; at worst, a sentence is reported as incomplete or passed through as unrecognized text.

IV. RESULTS AND EVALUATION

A. Experimental Setup

The parser was evaluated on a raw NMEA capture taken from the wristband’s LC76G module during a stationary outdoor field test. The capture spans approximately 131 seconds (13:07:44.622 to 13:09:55.359, capture-local time) and contains 1343 raw log lines: 1300 were dispatched as NMEA

TABLE I
NMEA SENTENCE COUNTS, 131-SECOND FIELD CAPTURE

Type	Description	Count
GGA	Global positioning fix	89
RMC	Recommended minimum navigation info	90
GSA	Active satellites / DOP	357
GSV	Satellites in view	573
VTG	Course and speed	90
GLL	Geographic position (legacy)	89
Other	Proprietary / unrecognized	12
Total		1300

sentences (1288 decoded into structured fields, 12 proprietary or unrecognized but still well-formed, see Table I), 18 were blank and produced no output, and 25 lacked a leading \$ character entirely and were written verbatim under a non-NMEA label, discussed in Section IV-D.

B. Sentence-Type Distribution

Table I summarizes the per-type counts produced by the parser’s running counters. GSV dominates the distribution, consistent with the fact that satellites-in-view data for four active constellations (GPS, GLONASS, Galileo, BeiDou) is each split across one or more messages per epoch, while GGA, RMC, VTG, and GLL each appear once per epoch.

C. Raw-to-Parsed Comparison

Fig. 2 pairs two raw NMEA sentences exactly as captured from the LC76G module with the parser’s corresponding structured output, for a GGA fix and the first message of a GPS GSV block. The reported latitude and longitude in the GGA block match the value obtained by applying Eq. (1) to the raw `dclmm.mmmmm` fields by hand, confirming correct coordinate conversion, and each `Sat #n` entry in the GSV block maps directly onto one repeating PRN/elevation/azimuth/SNR quadruple in the raw, comma-delimited sentence.

D. Robustness to Malformed Input

Of the 25 non-NMEA lines, one is a genuinely malformed sentence missing its leading \$GNGSA, formatter, beginning instead with `,A,3,...,1.05,0.72,0.77,3*02` – most likely a dropped byte at the module-to-host UART boundary at the start of the capture. The remaining 24 are different: at 13:08:49, partway through the capture, the LC76G module’s own bootloader trace appears mid-stream

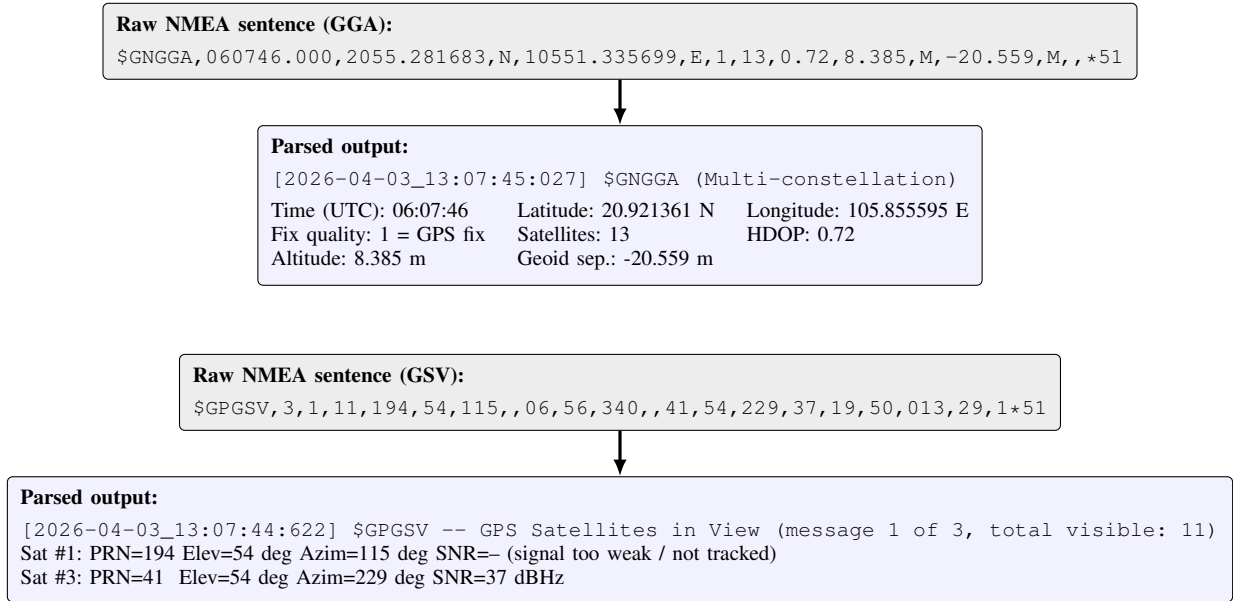


Fig. 2. Raw-to-parsed comparison for a GGA fix sentence and the first message of a GSV satellite-in-view block, both taken directly from the field capture. Each gray box is the literal raw sentence; each blue box is the parser’s corresponding structured output. The GGA fields are arranged in a grid here to save vertical space; the program itself prints one field per line.

– 22 lines of low-level boot output (register dumps, clock-frequency reports, flash initialization, and the line `Leaving the BROM`), evidently from the module restarting during the test – plus two manually inserted section markers (`acqui state`, `tracking state`). Because `extract_parts` requires a literal `$` to recognize a sentence, none of this was ever at risk of being read as NMEA data: all 25 lines fell through to the same non-NMEA passthrough path and were written verbatim rather than causing a crash or a misparse – the behavior anticipated by the design in Section III-B, now exercised against a real module reboot, a more demanding case than a single dropped-byte line alone.

E. Multi-Constellation Behavior

The capture confirms expected multi-constellation structure: GSA sentences appear in groups of up to four per epoch (one per tracked constellation, Section III-F), and GSV blocks for GPS, GLONASS, Galileo, and BeiDou are each present, with GPS contributing the largest share (11 satellites across 3 messages per epoch) and Galileo and BeiDou contributing few or none for most of the capture, consistent with a partial-sky test location. Combined GNRMC, GNVTG, and GNGGA sentences report a single fused position per epoch, with latitude and longitude drifting by under 0.00002° between fixes and altitude varying by a few meters – attributable to ordinary receiver noise at a stationary point rather than a parsing error, since GLL and RMC independently agree with the corresponding GGA fix.

V. CONCLUSION

This report described a small, dependency-free C parser for NMEA 0183 logs, developed to support field validation

of the GNSS subsystem on EDABK Lab’s Screenless Smart Wristband. The parser tokenizes raw sentences, converts coordinates to decimal degrees, maps talker IDs to constellation names, and produces a structured, human-readable report for six core sentence types. I evaluated the parser on a real 131-second, four-constellation field capture containing 1300 parsed sentences; it correctly decoded position, fix-quality, and satellite-visibility data, and passed through every piece of non-NMEA content encountered – a dropped-byte fragment and an unplanned mid-capture reboot of the GNSS module – without crashing or producing corrupted output. As the wristband project continues, this tool can be extended with fix-quality alerting thresholds or sentence-timing analysis to further support GNSS debugging on a device with no on-board display. Source code and field-test data are available at <https://github.com/Simbashrek276/NMEA-Data-Parsing-for-Screenless-Smart-Wristband-EDABK-Lab>.

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